Physics, Grade 11, SPH3U
Practice Test

Time: 2 hours
Total Marks: 110 **Score: _____ ÷ 110 × 100 = _____%**

Instructions

- The final test pages are numbered 1 through 21. Check to see that all 21 pages are attached. Inform the Final Test supervisor immediately if there are any pages missing.
- You may use a scientific calculator. There is a formula sheet provided, including a reference section with common constants.
- Write your answers in the space provided using pen or dark pencil.
- There are four (4) parts. An approximate time is given for each part. Look over the test before you begin, and leave some time to review your work at the end.

Part	Category	Marks	Time (min)
Preview			5
A	Waves and sound	23	24
B	Motion and forces	39	38
C	Energy and society	24	24
D	Electricity and magnetism	24	24
Review			5
Total		**110**	**120**

- This practice test is available to prepare for the paper or Microsoft Teams versions of the course final test. For the digital practice test, refer to your course.
- Please note that, for security reasons, marked tests are not returned to students.

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Formula Sheet – Physics SPH3U

Waves and sound

$f = \frac{\#\ cycles}{t} = \frac{N}{t}$

$f = \frac{1}{T}$

$v = f\lambda$

$f_{beat} = |f_2 - f_1|$ and $f_2 = f_1 \pm f_{beat}$

$v_{sound} = 332\ \mathrm{m/s} + \left(0.59\ \frac{\mathrm{m}}{\mathrm{s°C}}\right)T$

Doppler effect: $f_2 = f_1\left(\frac{v_{sound}}{v_{sound} \pm v_{source}}\right)$

Closed air column: $L_1 = \frac{\lambda}{4}$, $L_2 = \frac{3\lambda}{4}$, $L_3 = \frac{5\lambda}{4}$, etc.

Open air column: $L_1 = \frac{\lambda}{2}$, $L_2 = \lambda$, $L_3 = \frac{3\lambda}{2}$, etc.

Motion and forces

$\vec{v} = \frac{\vec{d}}{t}$

$\vec{a} = \frac{\overrightarrow{v_2} - \overrightarrow{v_1}}{t}$

$\vec{d} = \frac{\overrightarrow{v_1} + \overrightarrow{v_2}}{2}t$

$\vec{d} = \overrightarrow{v_1}t + \frac{1}{2}\vec{a}(t^2)$

$\vec{d} = \overrightarrow{v_2}t - \frac{1}{2}\vec{a}(t^2)$

$v_2^2 = v_1^2 + 2ad$

$\overrightarrow{F_{net}} = m\vec{a}$

$F_f = \mu F_N$

$\sin\theta = \frac{opposite}{hypotenuse}$ $\cos\theta = \frac{adjacent}{hypotenuse}$ $\tan\theta = \frac{opposite}{adjacent}$

If $ax^2 + bx + c = 0$, then $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

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Energy and society

$W = Fd$

$P = \frac{W}{t} = \frac{\Delta E}{t}$

$E_g = mgh$

$E_k = \frac{1}{2}mv^2$

$E_T = E_g + E_k$

$\%\ efficiency = \frac{useful\ output\ energy}{input\ energy} \times 100\%$

$E = mc^2$

$Q = mc\Delta T$

$Q = mL$

$Q_{lost} + Q_{gained} = 0$

Electricity and magnetism

$Q = Ne$

$I = \frac{Q}{t}$

$V = \frac{\Delta E}{Q}$

$E = VIt$

$V = IR$

$P = IV = I^2R = \frac{V^2}{R}$

$Cost = rate \times energy\ consumed$

$\frac{V_s}{V_p} = \frac{I_p}{I_s} = \frac{N_s}{N_p}$

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Physical constants

$g = 9.8\ \mathrm{m/s^2} = 9.8\ \mathrm{N/kg}$

$G = 6.67 \times 10^{-11}\ \frac{\mathrm{Nm^2}}{\mathrm{kg^2}}$

$c = 3.0 \times 10^{8}\ \mathrm{m/s}$

$e = 1.6 \times 10^{-19}\ \mathrm{C}$

Physical properties

Material	Specific heat capacity (J/kg°C)	L_f (J/kg)	Melting/ freezing point (°C)	L_v (J/kg)	Boiling/ condensation point (°C)
Ice	2.1×10^3	3.3×10^5	0	2.3×10^6	100
Water	4.2×10^3	3.3×10^5	0	2.3×10^6	100

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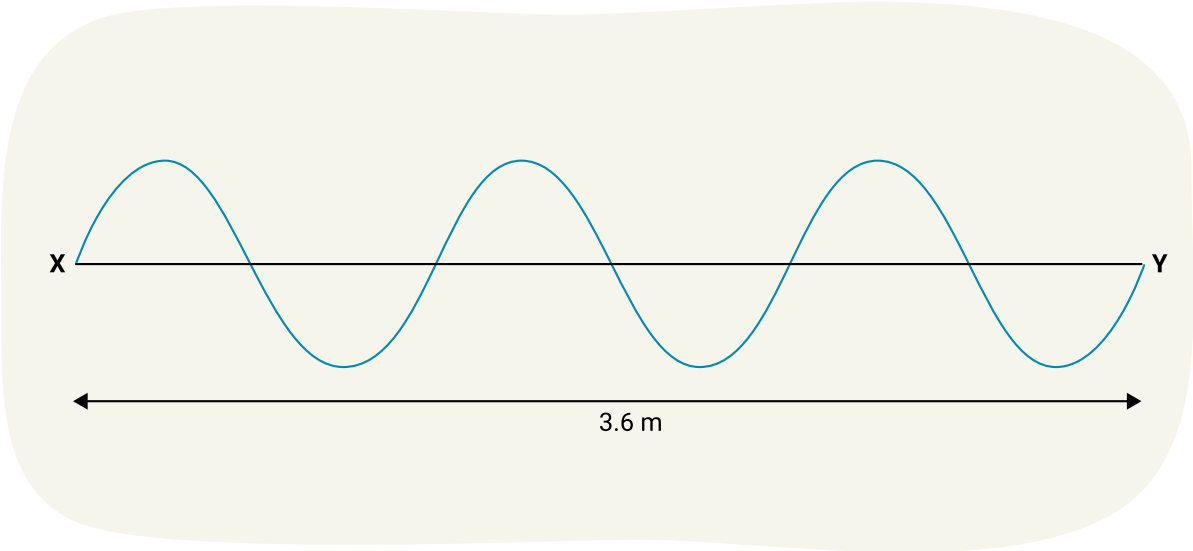
Part A: Waves and sound (23 marks) (approximate time 24 minutes)

1. A source makes 10 cycles in 5.0 s in a string. A diagram of some of the waves in the string is shown below. Complete the questions that follow.



a) What type of wave does the source make? How do you know? **(2 marks)**

b) Label the following measurements on the diagram: wavelength, amplitude. **(1 mark; ½ mark each)**

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c) Find the wavelength in metres. **(1 mark)**

d) Find the frequency. **(1 mark)**

e) Find the speed of the wave. **(1 mark)**

2. A car moving at 32.4 m/s sounds a 481 Hz horn when the temperature of the air is 24.0 °C. A person at the side of the highway hears the horn as the car is approaching and while it is moving away.

a) Why will the two frequencies be different? **(2 marks)**

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b) Calculate the observed frequency when the car is moving towards the person and away from the person. Indicate which frequency is the approaching frequency, and which is the frequency observed when the car is moving away. **(5 marks)**

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3. A source with a frequency of 60.00 Hz is used to make waves in a rope, which is 3.000 m long. It takes 0.1000 s for the waves to travel from one fixed end of the rope to the other. How many loops are in the standing wave on the rope? **(5 marks)**

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4. A 560 Hz tuning fork is held near the top of a closed air column while the water level, which is filled to the top, is slowly lowered. The temperature in the room is 12.0 °C. Find the first two resonant lengths in the air column. **(5 marks)**

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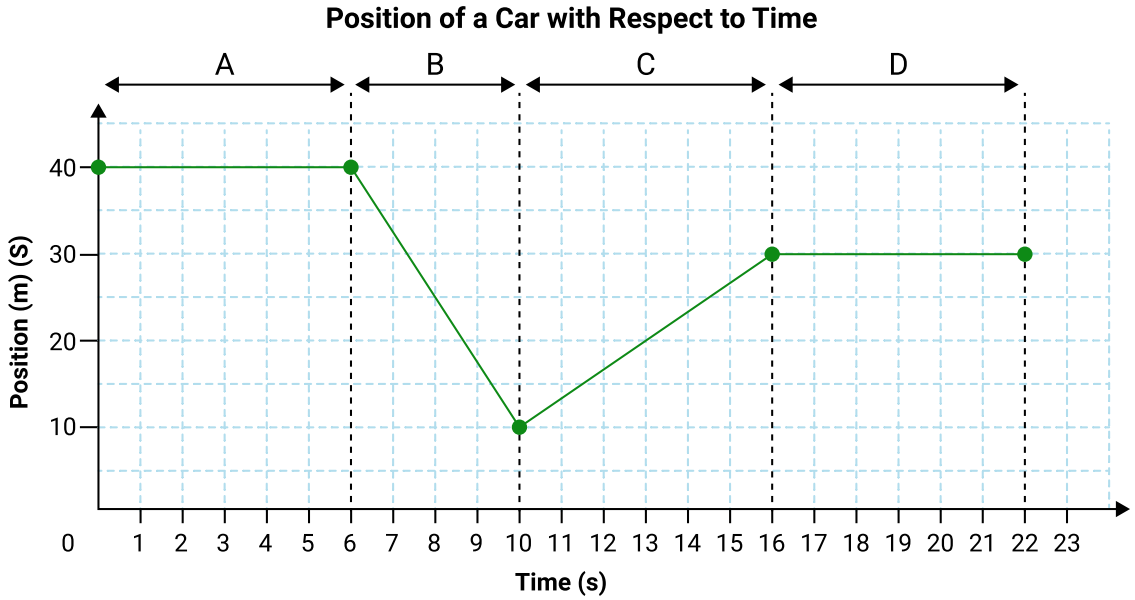
Part B: Motion and forces (39 marks) (approximate time 38 minutes)

5. Examine the following graph and then answer the questions that follow.



a) What is the total distance travelled? **(1 mark)**

b) What is the car's total displacement **(1 mark)**

c) What is the car's velocity in interval A? **(1 mark)**

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d) What is the car's velocity in interval C? **(2 marks)**

e) What is the car's average velocity over the first 16 seconds? **(2 marks)**

f) What is the car's average speed over the first 16 seconds? **(2 marks)**

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6. A boy in a wagon starts at rest and begins rolling down a straight ramp, gaining speed with uniform acceleration as he travels. In 6.00 s, he is travelling at 4.25 m/s.

a) What is his velocity at 10.0 s? **(4 marks)**

b) How long will it take him to reach a speed of 28.0 m/s? **(2 marks)**

7. A golfer hits a ball, giving it a velocity of 80.0 m/s [10° above horizontal].

a) How long, in seconds, is the ball in the air for before it hits the ground? **(4 marks)**

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b) Find how far the ball travels horizontally before landing on the ground. **(3 marks)**

8. A hockey puck with a mass of 0.168 kg is sliding on ice when it hits a rough patch that exerts a frictional force of 0.340 N [S].

a) Draw the free-body diagram of the puck while it slides on the rough patch. **(1 mark)**

b) Determine the puck's acceleration as it slides on the rough patch of ice. **(3 marks)**

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9. A child's wagon with rubber base is being pulled along the street. The wagon and its load have a combined mass of 28.0 kg. The coefficient of friction for rubber on dry asphalt is 1.07.

a) Find the force of friction. **(2 marks)**

b) How large must the pulling force be in order to give the wagon an acceleration of 0.50 m/s² [forward]? **(3 marks)**

c) If the wagon starts from rest, find its displacement after 22.0 s. **(2 marks)**

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10. A passenger train has six identical cars with a mass of 1500 kg each. They are accelerating at 0.80 m/s² [N]. Calculate the tension in the coupling between car 4 and 5. **(2 marks)**

11. A parcel is dragged with an applied force of 52 N [W] on a rough surface. The force of friction on the parcel is 52 N [E]. Describe the motion of the parcel. **(2 marks)**

12. Use Newton's laws to explain why a cannon rolls backwards after shooting a cannonball forward. **(2 marks)**

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Part C: Energy and society (24 marks) (approximate time 24 minutes)

13. A child with a mass of 43.0 kg is on a swing. At maximum amplitude, the swing is 2.60 m above the ground.

a) Find the child's gravitational potential energy at a height of 2.60 m. **(2 marks)**

b) How fast is the child moving when she is 1.25 m above the ground? **(3 marks)**

14. For each of the following situations, state whether work is being done or not. Explain how you know.

a) A student wearing a backpack walks across the room? **(2 marks)**

b) A student pushes a snow shovel across a driveway. **(2 marks)**

c) The brakes of a car are applied as the car slows to a stop. **(2 marks)**

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15. A piece of ice with a mass of 65.0 g is transferred directly from the freezer, at a temperature of -7.0 °C, to an insulating cup containing 200.0 g of water at 75 °C. Calculate the final temperature of the water. Ignore any heat lost to the cup and its surroundings. **(6 marks)**

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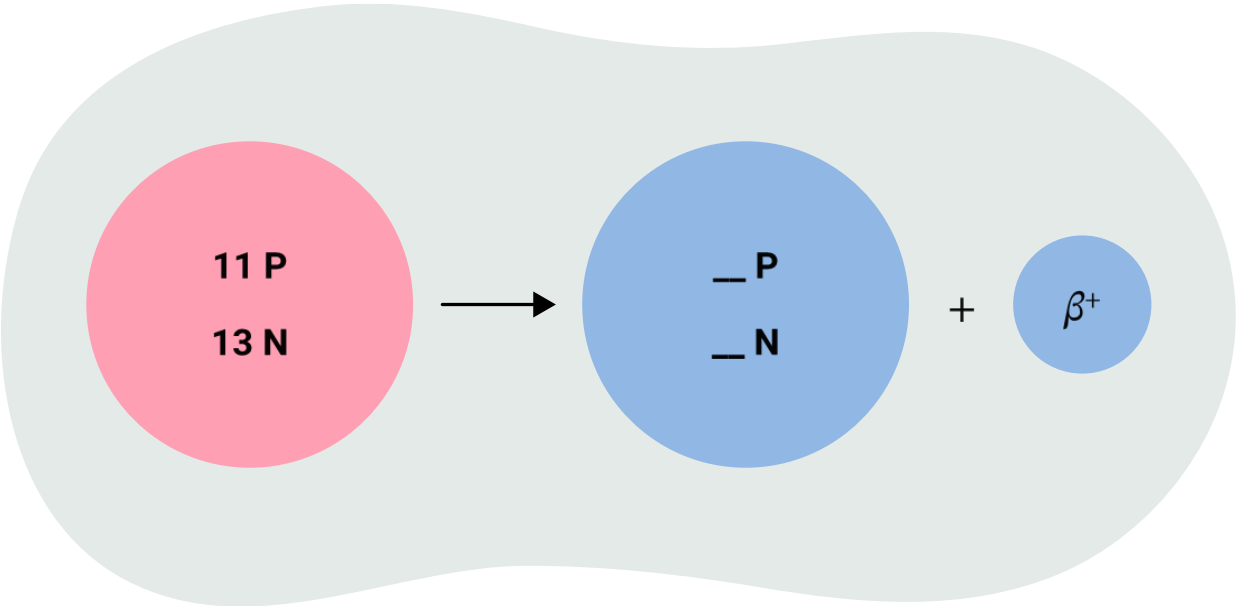
16. Carbon-14 has a half-life of 5,730 years. A sample of carbon-14 has a mass of 2.0×10^{-6} g.

a) How much carbon-14 will be left after 22,920 years? **(3 marks)**

b) Why can carbon-14 be used to date things that have died, but not to date rocks that formed many millions of years ago? Give two reasons. **(2 marks)**

17. Consider this nuclear reaction as the parent atom undergoes beta-positive decay. How many protons and how many neutrons does the resulting nucleus have? **(2 marks)**



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Part D: Electricity and magnetism (24 marks) (approximate time 24 minutes)

18. An electric kettle with a current of 8.0 A is connected to a 120 V source for 2.0 minutes.

a) How much energy does the kettle use? **(2 marks)**

b) What is the resistance of the kettle? **(2 marks)**

c) Find the charge (in coulombs) that flows through the kettle to make this current. **(1 mark)**

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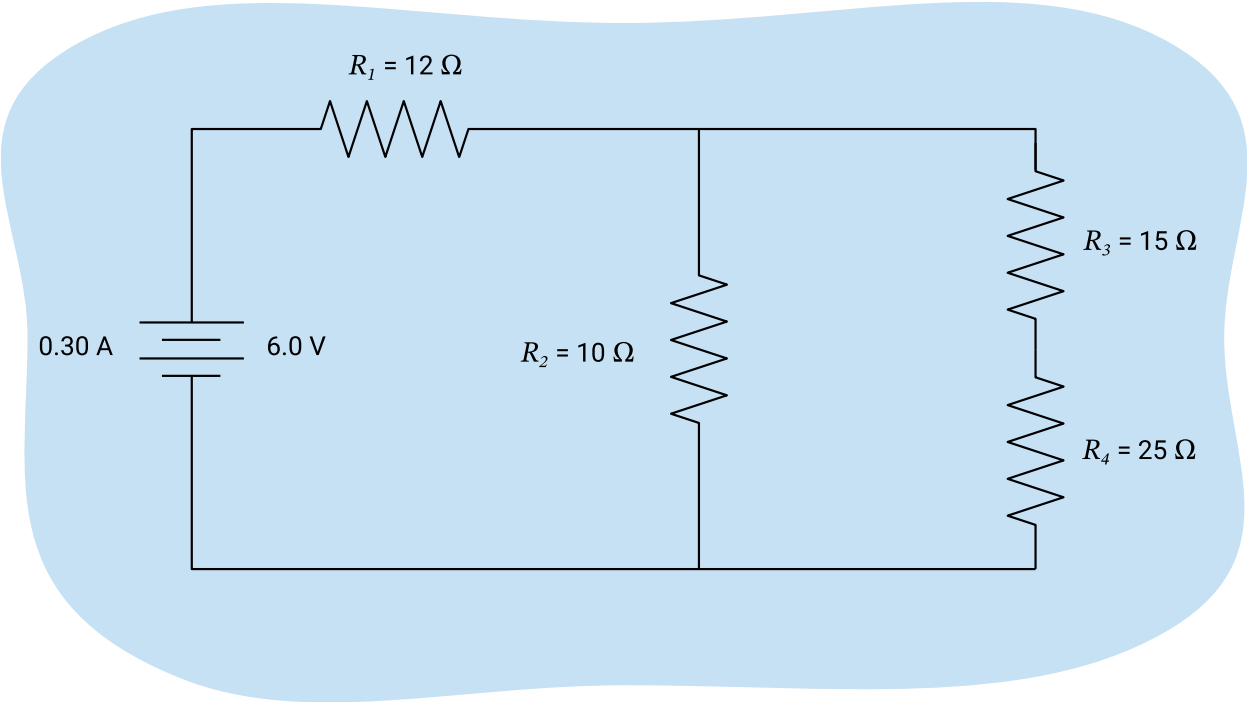
19. Examine the following circuit diagram with four resistors connected to a battery.



Find all of the missing values. Use the table that follows to organize your answers. **(12 marks)**

	Resistance (Ω)	Current (A)	Voltage (V)
Total (T)			
Resistor 1			
Resistor 2			
Resistor 3			
Resistor 4			

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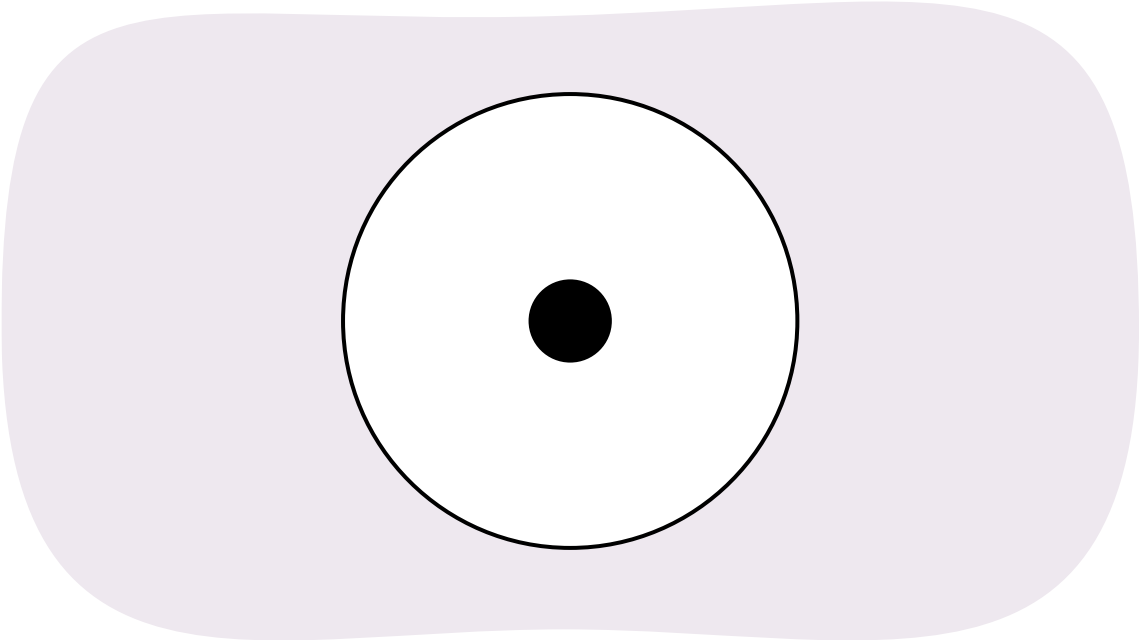
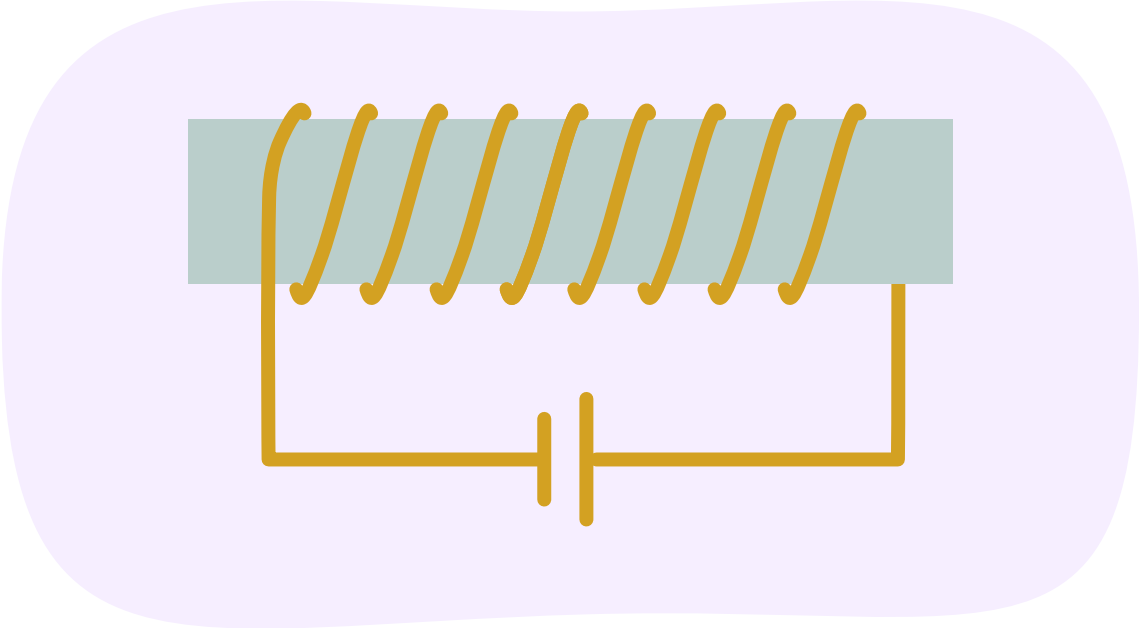
20. Draw the magnetic field lines around the following conductors.

a) This straight conductor. **(1 mark)**



b) This coil. **(1 mark)**



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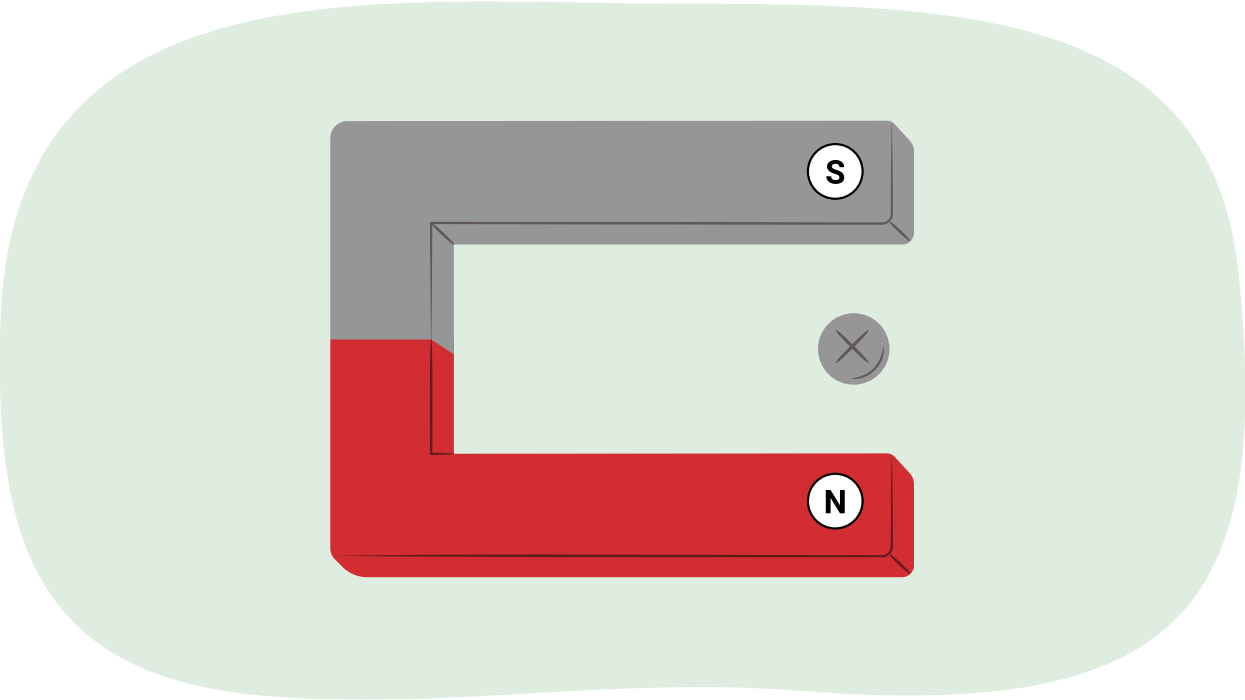
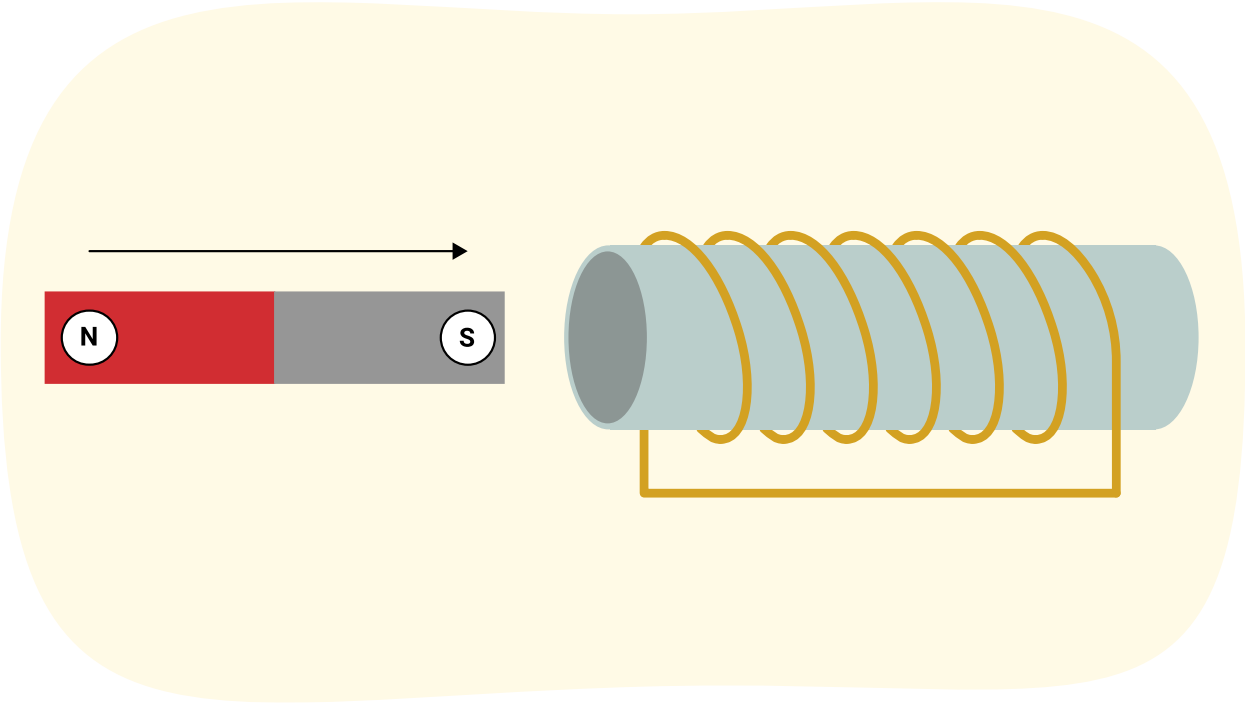
21. Complete the following diagram by drawing both magnetic fields and indicating the direction of the magnetic force on the conductor. **(2 marks)**



22. Describe the magnetic poles of the coil and draw the magnetic field that is induced when the following permanent magnet is moved toward the coil. **(2 marks)**



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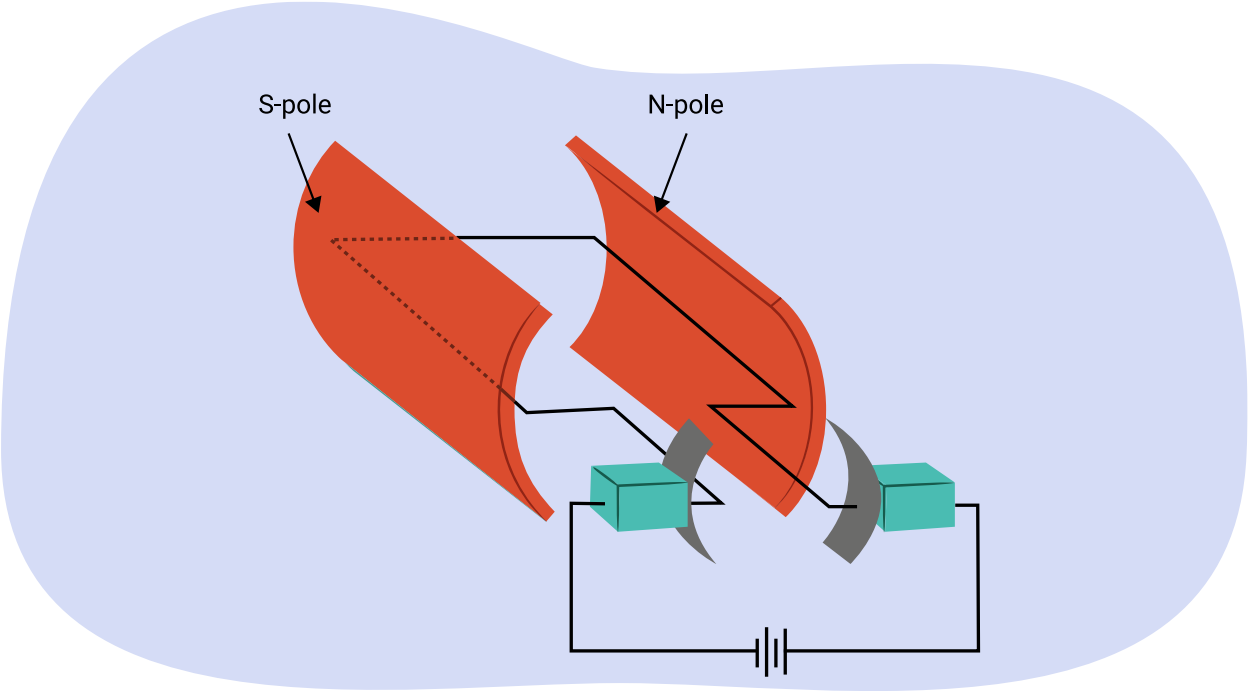
23. Describe which way the coil will rotate and explain how you know. **(1 mark)**



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